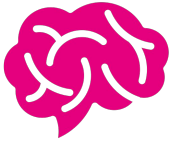


Solubility and temperature

Lesson objectives

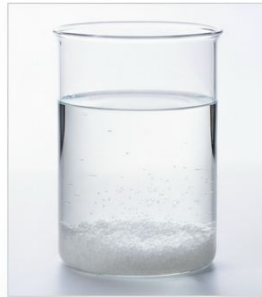
- To plan an investigation to test how temperature affects solubility.
- To collect data accurately and record it in a table.
- To draw and interpret a graph.



A



B



C

Which beaker shows the **solution**?
Where is the **solvent**?
Which one shows **undissolved solute**?

Keywords: solubility, temperature, variable, fair test, data

Success Criteria

Working towards

I can follow instructions to test how temperature affects solubility.

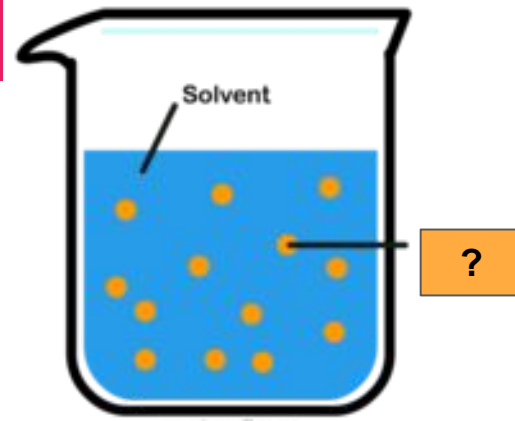
Working at

I can record results and draw a graph.

Working beyond

I can plan, carry out, and interpret results in relation to the effect of temperature on solubility.

Solubility and Temperature



1. What is the missing label from the diagram?

Solute

2. If a solution is not able to dissolve any more solute, the solution is

saturated

3. If the solution is able to dissolve more solute, the solution is

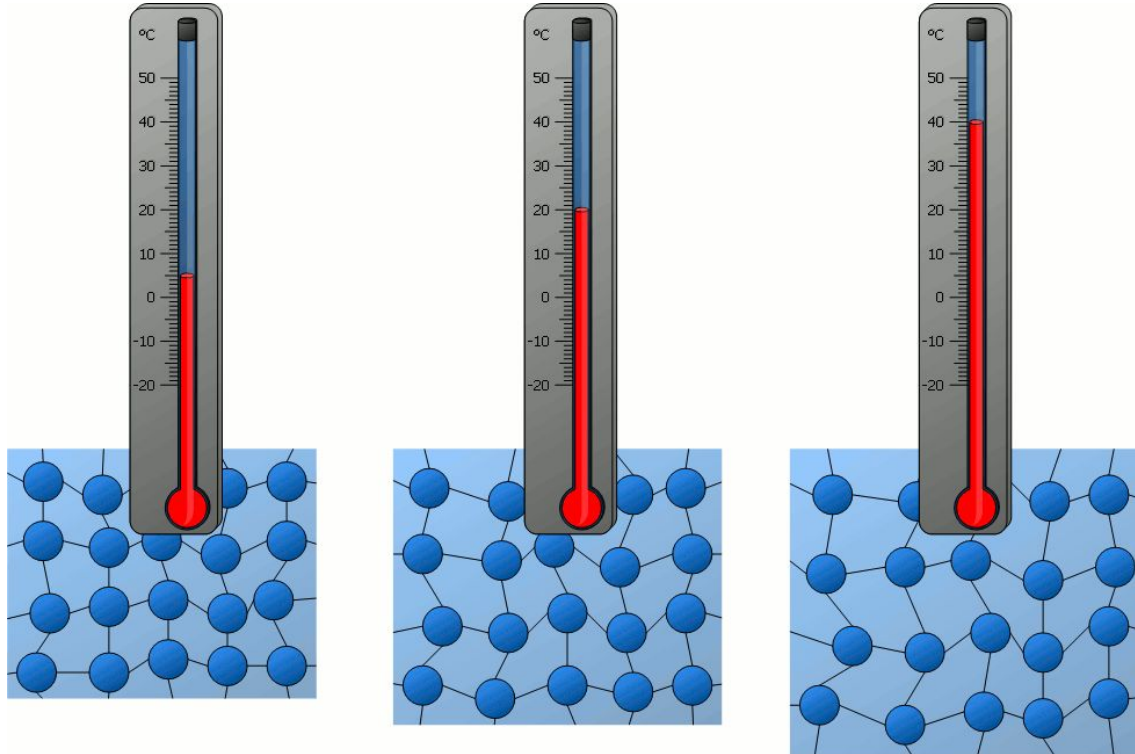
unsaturated

4. If we add more water to the solution, the solution becomes more

dilute

Solubility and Temperature

When we increase the temperature of water, there will be more thermal energy, so more kinetic energy



How will this affect solubility?

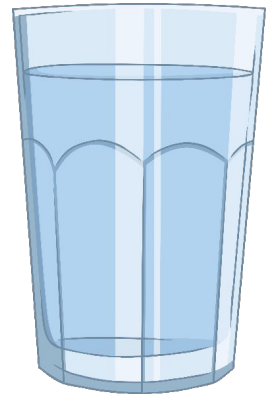
Solubility and Temperature

25:00

You are going to plan a safe and valid experiment into the solubility of sugar or salt (the solute) in water.

Solubility is a measure of how much solute will dissolve.

You will record how much sugar dissolves at different temperatures.



Planning an Experiment

Use the activity sheet to plan your experiment, the following slides will guide you

Planning an Experiment into Solubility

1. Background Information

Use the space below to write an introductory paragraph about solutions.

Words to use: solution, saturated, solute, solvent, solubility.

2. Method

Use the space below to write a clear method for adding 10g of solute at a time to a 25ml beaker of room temperature water until it is fully saturated. Then repeat with different temperatures of your choice.

3. How the Particles Look Inside

Draw two beakers showing the particles inside of them.

Beaker A (water only)

Beaker B (water and solute)

4. Risk Assessment

Complete the table to consider any safety issues in this practical. The first one has been given as an example.

Hazard	Harm	How You Will Prevent Injuries
glass beaker	Could break and cut skin	Keep floor clear and hold beaker securely; place in middle of table, wear goggles.

5. Variables

a. The independent variable is _____

b. The dependent variable is _____

c. Some control variables are _____

d. How we will monitor them: _____

6. Prediction

I predict that _____

I think this because (What is the science behind this idea?) _____

25:00

Solubility and temperature

Lesson objectives

- To conduct an investigation to test how temperature affects solubility.
- To collect data accurately and record it in a table.
- To draw and interpret a graph.

Keywords: solubility, temperature, variable, fair test, data

Background Information



Words to use: solution, saturated, solute, solvent.

When a substance is dissolved into a liquid, we call the substance dissolving a s _____ , the liquid a s _____ , and the mixture formed is called a s _____ .

We are going to plan an experiment to see how much solute dissolves at different temperatures, this is known as solubility. If we keep adding the solute and no more dissolves, then the solution is fully s _____ .

Method: put these steps in the correct order

- a. Add 10g of sugar (solute) at a time, noting down how much you have added.
- b. Stir gently with a stirring rod, applying the same amount of force each time.
- c. Clear a suitable working area, tie long hair up and put on safety goggles.
- d. Fill a glass beaker with 25cm^3 of water (solvent) at room temperature.
- e. Repeat steps b to d with hotter water. The temperatures we will use are 50°C and 100°C .
- f. Keep adding the solute until no more dissolves and the solution is fully saturated and record the total amount of grams of solute added.

Method

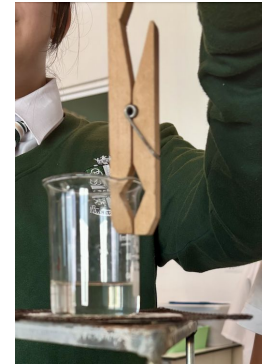
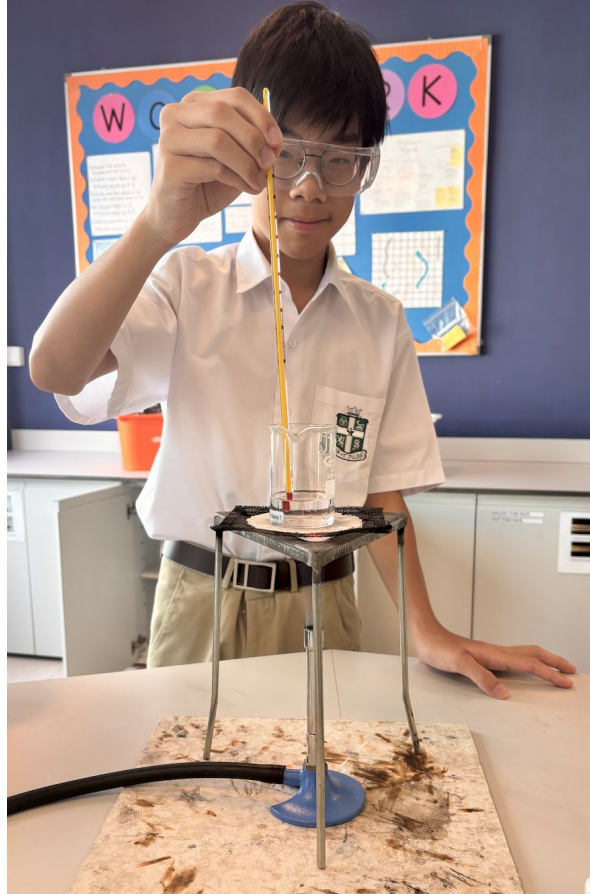
1. Clear a suitable working area, **tie** long hair up and put on safety goggles.
2. Fill a glass beaker with 25cm^3 of **water** (solvent) at room temperature.
3. Add 10g of **sugar** (solute) at a time, noting down how much you have added.
4. Stir gently with a **stirring rod**, applying the same amount of force.
5. Keep adding the solute until no more dissolves. Record the total amount of grams of solute added.
6. Repeat steps 2 to 5 with hotter water. The temperature should be 100°C .

TIP:

- Use an appropriate sized beaker so that the bulb of the thermometer can be covered by the water.

Equipment

- 3 beakers
- Mass balance
- Weighing boat
- Spatula
- Stirring rod
- Bunsen burner
- Tripod
- Gauze
- Thermometer
- Test tube holder
- Water
- Sugar



Hypothesis

I hypothesise that as the temperature of the water increases, (more/ less) sugar will dissolve.

Results

What you change - temp

What you measure – amount of sugar

What you keep constant - stirring

Temperature (°C)	Mass of sugar dissolved (g)

Investigating the effect of temperature on solubility



25°C



50°C



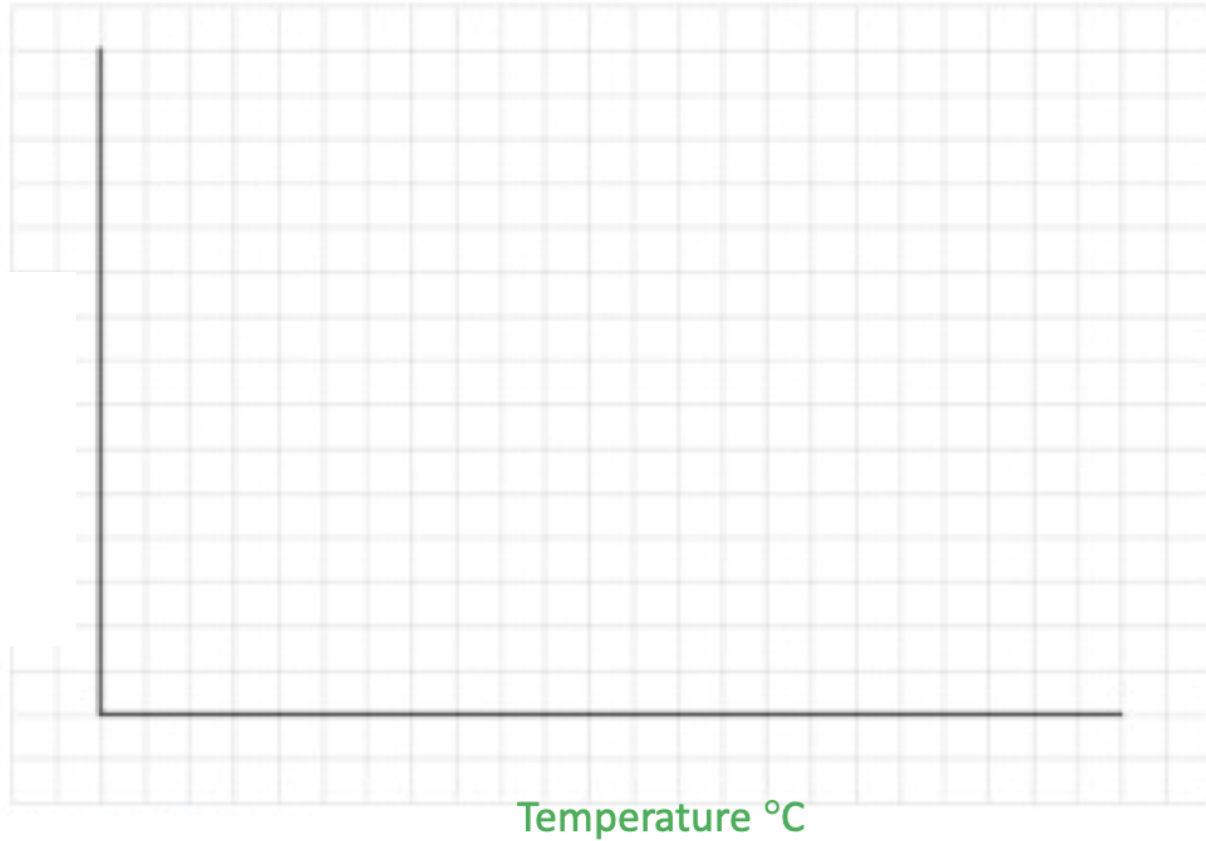
100°C



COUNTDOWN
40 00
MIN
with SIMPLE BEEPS

Results

Mass of
sugar
dissolved
(g)

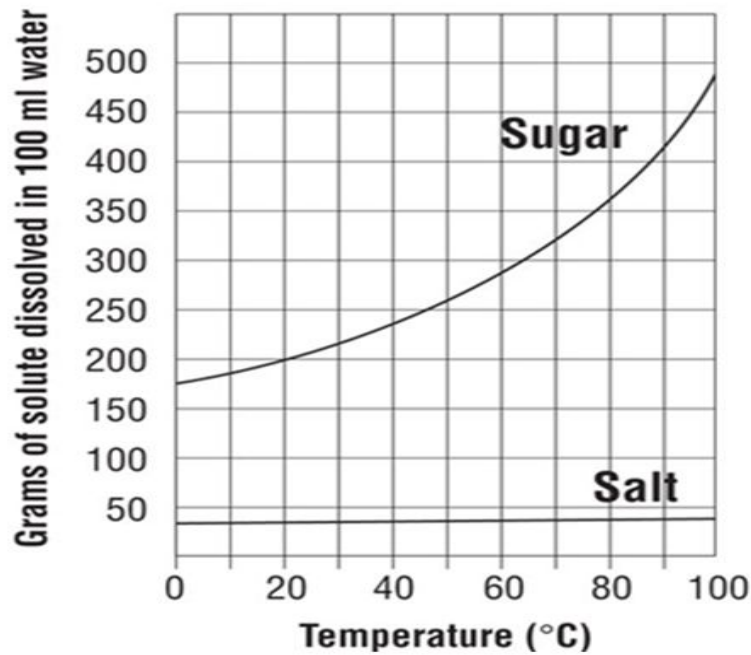


Conclusion

Conclusion:

Conclusion

As the temperature of the water increases, the amount of sugar that can dissolve before the solution becomes saturated increases, this is due to there being more kinetic energy in the water



1 level teaspoon $\approx$ 4 g

Compare your results with the results on the graph.

1. Were your results the same/ different? - why?
2. What do you think would have happened if you put the sugar in but didn't mix it?
3. Why do you think we used sugar instead of salt?